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Episode 52 · November 9, 2026 · 5 papers

Adaptive Hybrid Force-Vision Control with EMG-Derived References for Deformable Bottle Manipulation

SKIM

Jeffrey Lee et al.

A robot gripper squeezes lab wash bottles to dispense a target volume, switching between force and vision feedback, with grip force targets copied from a human's forearm muscle signals.

PROBLEM Squeeze bottles deform and then spring back, so a pure force controller loses track of how much the bottle has actually collapsed and dispenses the wrong amount. Lab automation work has mostly avoided this kind of deformable, compliant tool.

METHOD They build a hybrid controller that weights force feedback and camera-based deformation feedback differently depending on the contact phase, covering squeeze, hold and release. The force reference comes from human demonstrations, where EMG from the demonstrator's arm is mapped to a desired grip force profile. The robot then tracks that profile under the hybrid controller.

RESULTS Across 12 conditions, made of 2 target volumes, 2 bottles and 1 to 3 squeezes, the hybrid controller's mean absolute dispensing error was 6.54 mL. The pure force controller's error was 15.92 mL. The authors also report that the hybrid controller tracked the EMG-derived force profile more closely. The abstract gives no trial counts, no variance and no target volumes, so the relative error cannot be judged.

CAVEAT The only baseline is a pure force controller, which is a weak comparison. A vision-only controller and a fixed-weight hybrid are both missing, so we can't tell whether the gain comes from the adaptive switching or simply from adding vision. With only two bottles and an EMG-to-force mapping that probably comes from a single demonstrator, the setup is too narrow to show the method generalises.

WHY IT MATTERS This is outside your area. It is single-robot manipulation control with no vehicles, no communication and no negotiation between agents. The only loose link is the idea of switching feedback sources based on interaction state, and the cooperative adaptive cruise control literature already covers that better for your purposes. You can safely drop it.

arXiv:2611.07235v1 PDF incremental

A First Look at Starlink Maritime on Cruise Ships

SKIM

Jinwei Zhao, Kousar Malekinasab, Jianping Pan

The authors took a phone and Starlink dish telemetry onto two cruise ships and found that the ship's own router and per-passenger rate limits, not the satellite link, decide what passengers get.

PROBLEM LEO satellite performance has been measured on land, on moving cars and on aircraft, but not on cruise ships. On a cruise ship, several dishes sit behind a multi-WAN router that rations capacity among thousands of users, so nobody had separated the satellite's effect from the onboard network's.

METHOD This is a measurement study with no new system. It covers two ships, Utopia of the Seas near the Bahamas and Silver Muse in the Adriatic. It combines dish telemetry, tests run from a passenger device out to the internet, and probes from cloud servers near Starlink points of presence back toward the ship, so the onboard bottleneck can be told apart from the satellite path.

RESULTS The abstract gives no numbers. Its findings are qualitative: each passenger device is rate limited, the public IP address changes even when the serving point of presence stays the same, and the choice of point of presence follows neither the ship's maritime jurisdiction nor simple distance.

CAVEAT There are only two ships, one passenger vantage point each, and a single voyage window per ship, so the claims about the onboard network really describe two cruise lines' rate-limit policies, not Starlink maritime in general. The point-of-presence finding is only a hint, since the abstract offers no explanation and no controlled test of what drives the assignment.

WHY IT MATTERS This is outside your area. It has nothing on vehicle-to-vehicle negotiation, maneuver messages, platooning or joint trajectory planning, and its satellite backhaul setting is far from the sidelink latency budgets that maneuver coordination depends on. The only loose link is the general lesson that the access network's policy layer can matter more than the radio link, and you already know that.

arXiv:2611.07191v1 PDF incremental

cuRoboV2-Fast: Fast Torque-Limited Collision-Free Motion Planning

SKIM

Balakumar Sundaralingam, Adithyavairavan Murali, Stan Birchfield

NVIDIA made its GPU arm motion planner about fifteen times faster by switching to B-spline trajectories and adding a fast time-scaling step that respects joint torque limits under a changing payload.

PROBLEM Collision-free arm planners like cuRoboV2 were too slow for tight perception-to-action loops once payload-dependent torque limits were enforced. Separate time-parameterization tools like TOPPRA added latency and did not account for torque together with the collision optimization.

METHOD They optimize B-spline control points instead of dense waypoints, which needs fewer solver iterations, and they speed up the refinement step with GPU kinematics and differentiable inverse dynamics, so changing payloads come in through the dynamics. A second stage called TimeOpt keeps the path fixed and optimizes how fast it is traversed under sampled velocity, acceleration, jerk and torque limits. Because each spline segment depends only on a few neighbors, the Newton systems are banded and can be solved quickly on the GPU.

RESULTS With a three kilogram payload they report 4.4 milliseconds mean planning time on an RTX 5090, fourteen point seven times faster than cuRoboV2, and on a Jetson AGX Orin it beats cuRoboV1 plus TOPPRA running on the 5090. On a real YAM arm with stereo perception, mean pick-and-drop cycles took 5.93 seconds, thirty four point six percent shorter than cuRoboV2.

CAVEAT Every comparison is against the authors' own earlier cuRobo versions plus TOPPRA, so it is unclear how the method does against other fast optimizers or sampling planners. Torque limits are only checked at sampled points, not guaranteed between them. The hardware result is one arm on one pick-and-drop task, and the abstract says nothing about how often planning fails or how good the paths are.

WHY IT MATTERS This is not relevant to your work. It plans for a single robot arm, with no communication, negotiation or multiple agents. The only thing you could borrow is the engineering idea of banded spline systems for fast GPU time-scaling, which might help if you need to solve many vehicle trajectory problems in parallel, but you would not miss anything by skipping it.

arXiv:2611.06913v1 PDF incremental

Reinforcement Learning Enables Scaling Generalist Robot Policy Improvement

SKIM

Michal Korniak et al.

Fine-tuning a robot foundation model with online reinforcement learning on many manipulation tasks at once can improve all of them together, and value-based RL with the right mix of training tasks even raises success on tasks it never trained on.

PROBLEM Online RL fine-tuning of generalist robot policies has mostly been done one deployment task at a time. That throws away the multi-task breadth of the base model, and nobody had shown whether improvement on many tasks at once is possible or what recipe makes it work.

METHOD The authors build a structured set of evaluations for multi-task improvement during deployment. They then systematically compare design choices for running online RL on a pretrained generalist policy. These choices cover how to estimate value, how to extract a policy from that value estimate, and how to split data collection and compute across tasks.

RESULTS The abstract gives no numbers, no named benchmark and no named baselines. It reports only qualitative findings: multi-task online improvement works, it carries over to tasks held out from online training, and that carryover depends on which tasks are trained and on using value-based RL rather than alternatives the abstract does not name.

CAVEAT Calling this generalist improvement rests on evaluations the authors designed themselves. So the result depends on how far the held-out tasks really are from the training tasks, and the abstract does not say whether that is a new object, a new skill, or just a rephrased instruction. The abstract also never says whether the experiments ran on real robots or in simulation, or at what scale. The claim of being among the first is hedged and cannot be checked from the abstract.

WHY IT MATTERS This is peripheral to your work. It is about single-robot manipulation policies, with no multiple agents, no communication, no negotiation and no coordination protocol. The only possible link is a methodological idea if you ever fine-tune a learned driving or negotiation policy across many scenarios at once. Even then, the takeaway that value-based RL and a well-chosen training mix drive generalization is a general RL lesson you can get from the abstract.

arXiv:2611.06905v1 PDF notable

Dexterous Robotic Hands in the Foundation-Model Era: A Full-Stack Survey of Hardware, Data, Cross-Embodiment Transfer, Learning, and Evaluation

SKIM

Ji Pei

A single-author narrative survey of dexterous robot hands from 2024 to late 2026. It argues that human video data, shared action spaces and foundation policies still run into hand-specific limits on force, latency and contact, and that task success rates overstate how ready these systems are to deploy.

PROBLEM Work on dexterous hand hardware, learning from human data and evaluation is usually judged one area at a time. That makes it hard to tell whether progress in one layer actually carries over to a working system.

METHOD It is a structured narrative review, not a systematic review or a meta-analysis. It sorts the literature into five layers: hardware, data collection, cross-embodiment transfer, learning and foundation models, and evaluation. Human hand-object understanding and tactile sensing run across all five. It adds comparative tables, five case analyses and a system architecture, and it flags vendor specifications and company-reported results separately from controlled within-study comparisons.

RESULTS There are no pooled quantitative results, and the author says outright that it is not a performance ranking. The output is three qualitative conclusions, eleven open problems and a proposal for standard reporting formats. The abstract gives no benchmark numbers.

CAVEAT One author covering a fast-moving field that leans heavily on company claims, with no systematic search protocol, means the coverage and the three conclusions could reflect what the author chose to include. The conclusions themselves are fairly expected points. Of them, only the warning about zero-shot contact-rich transfer between different hand designs is backed by an explicit statement about missing evidence.

WHY IT MATTERS This has nothing to do with your work. It is about manipulation by a single robot hand. There is no multi-agent coordination, no communication protocol and no negotiation between vehicles. The only loose link is the general point that a shared action representation does not remove platform-specific limits, which echoes heterogeneous vehicles in a platoon. That is not worth reading time.

arXiv:2611.06724v1 PDF incremental